

1. Write a Pandas program to select the distinct department id from employees file.

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
30	Purchasing	114	1700
40	Human Resources	203	2400
50	Shipping	121	1500
60	IT	103	1400
70	Public Relations	204	2700
80	Sales	145	2500
90	Executive	100	1700
100	Finance	108	1700
110	Accounting	205	1700
120	Treasury	0	1700
130	Corporate Tax	0	1700
140	Control And Credit	0	1700
150	Shareholder Services	0	1700
160	Benefits	0	1700
170	Manufacturing	0	1700
180	Construction	0	1700
190	Contracting	0	1700
200	Operations	0	1700
210	IT Support	0	1700
220	NOC	0	1700
230	IT Helpdesk	0	1700
240	Government Sales	0	1700
250	Retail Sales	0	1700
260	Recruiting	0	1700
270	Payroll	0	1700

Code:

```
import pandas as pd
data = pd.read_csv('Csv/Lab1.csv')
distinct_id = data['DEPARTMENT_ID'].unique()
print("Distinct Department IDs:")
print(distinct_id)
```

Output:

```
Distinct Department IDs:
[ 10  20  30  40  50  60  70  80  90 100 110 120 130 140 150 160 170 180
 190 200 210 220 230 240 250 260 270]
```

2. Write a Pandas program to display the IDs of employees who have done two or more jobs in the past.

EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
102	13-01-2001	24-07-2006	IT_PROG	60
101	21-09-1997	27-10-2001	AC_ACCOUNT	110
101	28-10-2001	15-03-2005	AC_MGR	110
201	17-02-2004	19-12-2007	MK_REP	20
114	24-03-2006	31-12-2007	ST_CLERK	50
122	01-01-2007	31-12-2007	ST_CLERK	50
200	17-09-1995	17-06-2001	AD_ASST	90
176	24-03-2006	31-12-2006	SA_REP	80
176	01-01-2007	31-12-2007	SA_MAN	80
200	01-07-2002	31-12-2006	AC_ACCOUNT	90

Code:

```
import pandas as pd
data = pd.read_csv('Csv/Lab2.csv')
result = data.groupby("EMPLOYEE_ID").filter(lambda x: len(x) >= 2 ). sort_values
(by="EMPLOYEE_ID")
print("Employees who have done two or more jobs:")
print(result)
```

Output:

Employees who have done two or more jobs:					
	EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
1	101	1997-09-21	2001-10-27	AC_ACCOUNT	110
2	101	2001-10-28	2005-03-15	AC_MGR	110
7	176	2006-03-24	2006-12-31	SA_REP	80
8	176	2007-01-01	2007-12-31	SA_MAN	80
6	200	1995-09-17	2001-06-17	AD_ASST	90
9	200	2002-07-01	2006-12-31	AC_ACCOUNT	90

3. Write a Pandas program to display the details of jobs in descending sequence on job title.

JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
AD_PRES	President	20080	40000
AD_VP	Administration Vice President	15000	30000
AD_ASST	Administration Assistant	3000	6000
FI_MGR	Finance Manager	8200	16000
FI_ACCOUNT	Accountant	4200	9000
AC_MGR	Accounting Manager	8200	16000
AC_ACCOUNT	Public Accountant	4200	9000
SA_MAN	Sales Manager	10000	20080
SA_REP	Sales Representative	6000	12008
PU_MAN	Purchasing Manager	8000	15000
PU_CLERK	Purchasing Clerk	2500	5500
ST_MAN	Stock Manager	5500	8500
ST_CLERK	Stock Clerk	2008	5000
SH_CLERK	Shipping Clerk	2500	5500
IT_PROG	Programmer	4000	10000
MK_MAN	Marketing Manager	9000	15000
MK_REP	Marketing Representative	4000	9000
HR_REP	Human Resources Representative	4000	9000
PR_REP	Public Relations Representative	4500	10500

Code:

```
import pandas as pd
data = pd.read_csv('Csv/Lab3.csv')
result = data.sort_values(by="JOB_TITLE", ascending=False)
print(result)
```

Output:

	JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
11	ST_MAN	Stock Manager	5500	8500
12	ST_CLERK	Stock Clerk	2008	5000
13	SH_CLERK	Shipping Clerk	2500	5500
8	SA_REP	Sales Representative	6000	12008
7	SA_MAN	Sales Manager	10000	20080
9	PU_MAN	Purchasing Manager	8000	15000
10	PU_CLERK	Purchasing Clerk	2500	5500
18	PR_REP	Public Relations Representative	4500	10500
6	AC_ACCOUNT	Public Accountant	4200	9000
14	IT_PROG	Programmer	4000	10000
0	AD_PRES	President	20080	40000
16	MK_REP	Marketing Representative	4000	9000
15	MK_MAN	Marketing Manager	9000	15000
17	HR_REP	Human Resources Representative	4000	9000
3	FI_MGR	Finance Manager	8200	16000
1	AD_VP	Administration Vice President	15000	30000
2	AD_ASST	Administration Assistant	3000	6000
5	AC_MGR	Accounting Manager	8200	16000
4	FI_ACCOUNT	Accountant	4200	9000